

AMAR SINHA

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SUMMARY

Highly motivated and responsible individual with strong leadership and collaboration skills. Currently pursuing B.Tech in Computer Science Engineering (Data Science) at Heritage Institute of Technology, Kolkata. Passionate about learning, problem-solving, and contributing to meaningful projects through innovation and teamwork.

EDUCATION

Heritage Institute of Technology, Kolkata, India

2023 – 2027

B.Tech in Computer Science and Engineering (Data Science)

– 1st Sem – 7.15 2nd Sem – 7.95 3rd Sem – 8.87 4th Sem – 9.04 5th Sem – 8.66 6th Sem – 8.26

Methodist School, Dankuni, West Bengal, India

2021 – 2023

ISC (Class 12) – 85% ICSE (Class 10) – 93%

TECHNICAL SKILLS

Programming Languages: Python, R, JavaScript (Node.js, React.js), C, Java, SQL

Data Science Tools: R Shiny, Random Forest, Forecasting Models, Data Visualization, Machine Learning (GLM, Classification, Regression)

Web Development: ReactJS, NodeJS, ExpressJS, MongoDB, HTML, CSS

APIs & Cloud: AWS (EC2, S3), Google Gemini API, Microsoft Azure Translation API, REST APIs

Developer Tools: Git, GitLab, VS Code, Linux/Unix Systems

PROJECTS

Who Wins 2029 | R, Shiny, Random Forest, Plotly, GeoJSON

2025

- Built an ML-powered election prediction platform that forecasts state-wise Lok Sabha winners using Random Forest models trained on historical election datasets.
- Designed an interactive Shiny dashboard featuring GIS-based India maps, prediction visualizations, historical seat trends, and party-wise analytics using Plotly and GeoJSON.
- Implemented dynamic filtering and real-time prediction updates, enabling users to explore election outcomes across multiple election years.

AI Surveillance Platform | React, FastAPI, YOLOv8, ByteTrack, DeepFace, PostgreSQL

2026

- Built an intelligent surveillance system capable of detecting and tracking individuals in uploaded videos and live camera feeds using YOLOv8 and ByteTrack.
- Developed AI modules for restricted-zone intrusion detection, loitering analysis, and optional DeepFace-based face recognition, with real-time alerts and dashboard visualization.
- Engineered a scalable FastAPI backend and React frontend with PostgreSQL persistence, JSON fallback storage, and Docker-based local deployment.

Kolkata Metro Route Planner | JavaScript, HTML, CSS, Google Maps API

2025

- Developed an interactive route planner that allows users to visualize and find optimal routes between Kolkata Metro stations.

INTERNSHIP

Indian Oil Corporation Limited (IOCL)

2024

Intern – Comprehensive Study of Loyalty Program

Kolkata, India

- Conducted research on loyalty programs across industries and proposed strategies for improving customer acquisition.

Sarvagna Innovation Private Limited

July 2025 – August 2025

AI Intern – Artificial Intelligence (4 Weeks Online Internship)

- Gained practical experience in Artificial Intelligence, including neural networks, generative AI, and conversational agents.
- Developed a simple CNN model for image classification using TensorFlow.
- Created AI-generated music and multimedia content using tools such as Boomy, AIVA, and OpenAI APIs.
- Built a basic FAQ Chatbot using Google Dialogflow and explored multimodal AI applications.